



Re: Please see whether you could run Huang Cheng's GLaBoost onto GRAPE dataset

From Zhang, Jia <jiazhang@smu.edu>

Date Thu 9/17/2026 5:37 PM

To Han, Zeyu <zeyuhan@smu.edu>

External Retrospective Validation of Longitudinal Glaucoma Progression Using GlaBoost + GRAPE

Objective. Conduct a focused retrospective external validation using the public GRAPE longitudinal glaucoma dataset. The experiment should test whether simple integration of GlaBoost-derived evidence across multiple visits improves glaucoma progression detection compared with assessment based on the latest encounter alone.

This is **not** intended to develop a new longitudinal deep-learning model. Use the existing GlaBoost pipeline as the visit-level AI engine and add only a simple, prespecified temporal integration layer.

GlaBoost currently represents each observation as an independent glaucoma-classification instance, combining fundus-image features, structured features, textual features, and optional clinician-derived information. Its image representation uses a pretrained ResNet-152, while multimodal classification is performed using XGBoost.

1. Construct the GRAPE longitudinal cohort

Use **eye**, not patient, as the longitudinal unit. Retain eyes with at least **3 evaluable longitudinal visits** and sufficient information to generate the selected GlaBoost-compatible visit-level outputs.

For every eye, create an ordered sequence

$$E_i = \{V_{i1}, V_{i2}, \dots, V_{iT_i}\}, \quad 3 \leq T_i \leq 9,$$

with the actual visit dates/times retained so that changes can be normalized by follow-up duration.

Do **not** require modalities unavailable longitudinally in GRAPE merely to reproduce the full UTSW GlaBoost configuration. Instead, define before analysis a **GRAPE-compatible GlaBoost configuration** using modalities/features that can be consistently mapped. This is justified because GlaBoost itself was evaluated under different modality combinations.

Create a short mapping table before running experiments:

GlaBoost input	GRAPE counterpart	Longitudinally available?	Use?
Fundus image	GRAPE CFP	Determine	Yes if adequate
Image embedding	ResNet-152 from CFP	Derived	Yes
C/D or related fundus features	GRAPE corresponding field, if available	Determine	If compatible
IOP/clinical features	GRAPE fields	Yes if available	Consider
Free-text rim description	No direct equivalent unless present	Determine	Likely no
UTSW OCT RNFL/ONH/GCC features	No equivalent serial OCT	No	No

GlaBoost paper's private cohort uses OCT-report biomarkers across RNFL, ONH, and GCC, so those features should **not be assumed available longitudinally in GRAPE**.

Before modeling, report: number of patients; number of eligible eyes; total eligible visits; median visits/eye [IQR]; median follow-up months [IQR]; and availability of each selected modality.

2. Generate visit-level GlaBoost evidence

Apply the **same fixed GlaBoost-compatible pipeline independently to every eligible visit**:

$$V_{it} \xrightarrow{\text{GlaBoost}} S_{it},$$

where S_{it} is preferably the continuous glaucoma probability/risk score rather than a binary decision.

If feasible, also retain selected interpretable outputs/features:

$$Z_{it} = [S_{it}, z_{it}^{(1)}, \dots, z_{it}^{(K)}].$$

Useful features could include C/D-related or other fundus-derived evidence when available. GlaBoost's original feature analysis found substantial contributions from cup-to-disc ratio and rim thinning, among other variables.

Important: Do not allow information from later visits to enter the GlaBoost computation for an earlier visit. Each S_{it} must represent what the existing AI could have produced from that encounter alone.

3. Primary A/B experiment

The primary comparison must hold the underlying GlaBoost detector constant.

A. Latest-encounter assessment

Use only the last eligible encounter:

$$X_i^A = S_{iT}.$$

If a classifier/calibration layer is required:

$$P(Y_i = 1) = \sigma(\beta_0 + \beta_1 S_{iT}).$$

This represents conventional encounter-based AI.

B. Simple longitudinal integration

Derive a small, prespecified set of temporal features from the same GlaBoost outputs:

$$X_i^B = [S_{iT}, \Delta S_i, Slope_i, Mean_i, Persistence_i].$$

At minimum calculate:

$$\Delta S_i = S_{iT} - S_{i1},$$

and estimate

$$Slope_i = \frac{\text{Cov}(t, S)}{\text{Var}(t)}$$

using actual elapsed time rather than visit number.

Mean summarizes repeated evidence. Define Persistence prospectively, for example, the fraction of visits exceeding a threshold established from training data. Do not try many definitions and select the one that gives the best GRAPE result.

Use **logistic regression as the primary temporal integrator**:

$$P(Y_i = 1) = \sigma(\beta_0 + \beta_1 S_T + \beta_2 \Delta S + \beta_3 Slope + \beta_4 Mean + \beta_5 Persistence).$$

Keep this model intentionally simple. **Do not use LSTM, GRU, Transformer, Mamba, etc. for the primary experiment.** Although such models were evaluated in GlaBoost, they were classification baselines rather than longitudinal patient-trajectory models.

The scientific comparison must remain:

Same GlaBoost + latest visit

vs.

Same GlaBoost + longitudinal trajectory

4. Progression reference standards

Evaluate A and B independently against each of GRAPE's three predefined progression outcomes:

$$Y^{PLR2}, \quad Y^{PLR3}, \quad Y^{MD}.$$

Do not combine these into a new composite ground truth for the primary analysis.

Very importantly, document whether any GlaBoost input or derived temporal feature uses VF information that directly participates in constructing PLR2, PLR3, or MD-slope labels. If so, flag this as potential circularity.

Ideally, the primary predictor should rely on **non-VF visit-level AI evidence** when evaluating against VF-derived progression references. This gives us the cleaner interpretation:

longitudinal AI/image/clinical evidence → progression

evaluated against an independently defined longitudinal VF reference.

5. Leakage controls and statistical design

These controls are essential.

All splitting must occur at the **patient level**, not visit or eye level. If both eyes of a patient are included, both eyes must remain in the same partition.

No future visit can contribute to an earlier visit's GlaBoost score. Feature selection, normalization, threshold selection, temporal-model fitting, and hyperparameter selection must use training data only.

Because GRAPE is small and progression cases may be relatively uncommon, use **patient-grouped stratified cross-validation** if feasible rather than creating a small single holdout set. Predictions used for final performance calculations should be **out-of-fold predictions**.

Most importantly, A and B must use:

- exactly the same eligible eyes;
- exactly the same reference labels;
- the same cross-validation folds;
- the same underlying GlaBoost visit-level outputs.

Thus, the primary experimental variable is **availability of longitudinal information**.

6. Metrics and uncertainty

Use **balanced accuracy as the primary endpoint**, because progression/non-progression may be substantially imbalanced:

$$BA = \frac{Sensitivity + Specificity}{2}.$$

For each reference definition also report AUROC, AUPRC, sensitivity, specificity, and F1.

The key statistic is the paired improvement:

$$\Delta BA = BA_{Longitudinal} - BA_{Latest}.$$

Estimate **95% confidence intervals by patient-level bootstrap**, resampling patients rather than individual eyes so that two eyes from the same patient remain clustered.

Report:

$$\Delta BA_{PLR2}, \quad \Delta BA_{PLR3}, \quad \Delta BA_{MD}.$$

Do not emphasize p-values; effect sizes and CIs are more important for this preliminary study.

7. What I need returned

Please return one concise analysis package containing:

Cohort: [N patients], [N eyes], [N visits], median [X] visits/eye, median [Y] months follow-up, plus progression prevalence separately for PLR2, PLR3, and MD.

Data mapping: a one-page table documenting exactly which GRAPE variables were mapped into GlaBoost and which original GlaBoost modalities were unavailable.

Model specification: exact definition of St; exact temporal features; logistic-regression specification; CV procedure; handling of missing visits/modalities; and all leakage controls.

Primary results:

Reference	N eyes	Progression n (%)	Latest BA (95% CI)	Longitudinal BA (95% CI)	ΔBA (95% CI)
PLR2					
PLR3					
MD slope					

Also provide AUROC, AUPRC, sensitivity, specificity, and F1 in a supplementary table.

Finally, provide **one figure** with three paired comparisons—PLR2, PLR3, and MD—showing latest-encounter versus longitudinal balanced accuracy with 95% CIs.

Please also send the analysis code, fixed random seeds/environment information, and a short description of any cases excluded and why.

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From: Zhang, Jia <jiazhang@smu.edu>

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Subject: Please see whether you could run Huang Cheng's GLaBoost onto GRAPE dataset

Preliminary external validation of longitudinal glaucoma progression

assessment. Building on our published multimodal glaucoma cohort and validated visit-level AI capabilities, we conducted a focused retrospective external validation using the publicly available GRAPE longitudinal glaucoma cohort. Of 263 eyes with 3–9 longitudinal visits, [N] eyes met prespecified eligibility criteria, providing a median of [X] visits over [Y] months of follow-up. Progression was independently defined using GRAPE's three prespecified longitudinal visual-field criteria (PLR2, PLR3, and MD slope). Existing validated computational capabilities were applied independently at each eligible visit, and their outputs were integrated using prespecified temporal changes and trends, without persistent longitudinal representation or reasoning. Compared with assessment using the latest

encounter alone, longitudinal integration improved balanced accuracy from [XX] to [XX] for PLR2 (Δ = [XX], 95% CI [XX–XX]), from [XX] to [XX] for PLR3 (Δ = [XX], 95% CI [XX–XX]), and from [XX] to [XX] for MD-slope progression (Δ = [XX], 95% CI [XX–XX]). [If supported: Improvements were directionally consistent across all three progression definitions.] These results provide external retrospective evidence that longitudinal evidence adds value beyond single-encounter assessment for glaucoma progression detection, while leaving the central UG3 challenge unresolved: transforming episodic AI evidence into persistent, traceable, uncertainty-aware longitudinal clinical intelligence.

GRAPE — strongest fit for longitudinal retrospective validation

[GRAPE official dataset collection on Figshare](#)

GRAPE was explicitly created as a **longitudinal glaucoma follow-up dataset**. It contains **263 eyes from 144 subjects and 1,115 longitudinal visit records**, with **3–9 visits per eye**.

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